Predicting Formal Protest Petitions Against Housing Development

Evidence from Austin, Texas

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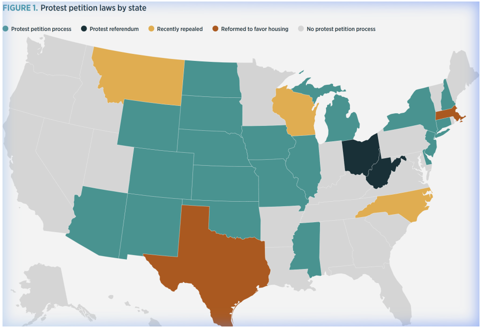
Housing Crisis and Local Opposition

The Affordability Problem

- Austin grew ≈40% between 2010–2024.
- New housing requires discretionary rezoning approval.
- Projects face public hearings and organized opposition.

The Political Mechanism

Texas law gave adjacent property owners a formal institutional veto: the **protest petition**.

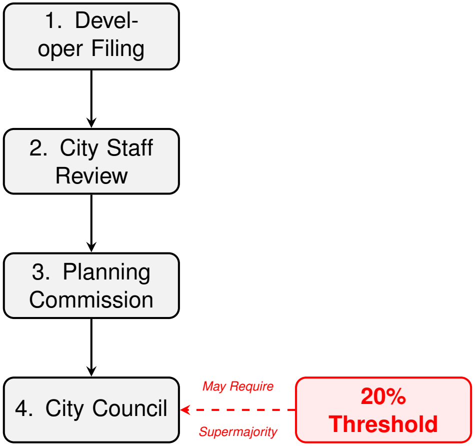


The Protest Petition: An Institutional Veto

Texas LGC Chapter 211: The 20% Rule

- Owners of 20% of land within 200ft may file written protest.
- A valid petition triggers a supermajority ($\frac{3}{4}$) council vote.
- Post-2014 council (11 members): 6 votes $\rightarrow$ 9 required.
- 21 states maintain similar petition statutes.

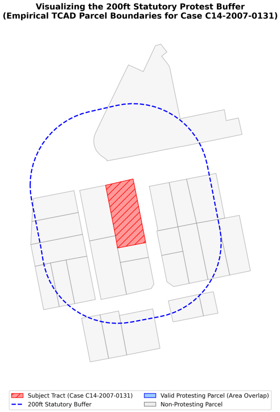
A localized minority can block citywide housing decisions.



Protest Petition Rules Across States

State	Threshold	Buffer	Supermajority
Texas	**20%**	**200ft**	**3/4**
Iowa	20%	200ft	3/4
New York	20%	100ft	3/4
Colorado	20%	100ft	2/3
Oklahoma	50%	300ft	3/5 or 3/4
Massachusetts	50%	300ft	2/3

Source: Furth & McKinley (2025). 21 states maintain active protest petition statutes.



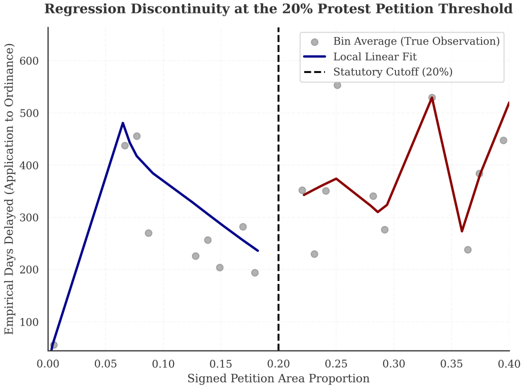
200ft parcel buffer defining the petition area

Why Petitions Matter: Institutional Costs

Reduced-Form Discontinuity at the 20% Threshold

- Running variable: reconstructed petition-share.
- Outcome: council processing time.
- Estimator: triangular-kernel WLS at the 20% margin.

Result: Crossing the threshold adds **42.5 days** (6.1 weeks).
95% CI: [26.4, 58.6] days. McCrary $p > 0.05$.



The Research Question

The Prediction Problem

Can we anticipate protest petitions *before* public hearings using only filing-date records?

Filing-Date Features

- Site geometry: lot area, height/FAR deltas.
- Current vs. requested zoning class.
- Buffer demographics and TCAD appraisals.
- Historical petition rates in the district.

A Sample Case

Feature	Case C14-2018-0156
Gross Site Area	4.92 Acres
Height Request	+15.5 ft
FAR Request	+0.43
Neighborhood	Riverside
Result	**Valid Petition**

Race/ethnicity excluded from predictive features; reserved for post-hoc equity audits only.

The Rarity Problem

Sample Characteristics

- $N = 6783$ cases, 2007–2024.
- Base rate: $\approx 2.6\%$ (1 in 38 cases).
- Test-set positives: $n_{\text{pos}} = 15$.

With so few events, confidence intervals are wide. The model output is a **score per case** (0–1); cases are ranked by that score, not classified by threshold.

Evaluation Design

- Train on years $\leq t$; evaluate on years $> t$.
- Temporal holdout for the test set; random cross-validation within training.

Primary metric: PR-AUC

AUROC omitted: we are predicting a rare class.

How We Arrived at $N = 6783$

Selection Step	Cases
Raw Austin open-data records	15,238
Discretionary cases only	7,153
Usable filing-year information	6783
Analytic sample	**6783**

Excluded: non-discretionary variances, ETJ cases (petition statute does not apply), and cases with unrecoverable timestamps.

Illustrative Panel Row

Case	Year	Lag Pet.	Med. Inc.	Own. Sh.
C14-2015-001	2015	3	$85k	62%
C14-2015-002	2015	0	$42k	21%
C14-2015-006	2015	7	$112k	85%
C14-2016-009	2016	1	$55k	34%
C14-2016-012	2016	4	$95k	71%

Each row is one case; all features are available at filing date.

Timeline Integrity: As-Of Features

Enforcing the Constraint

- 17 years of zoning records from Austin's open-data portal (2007–2024).
- Filing dates back-calculated from statutory minimums.
- American Community Survey (ACS) and Travis County Appraisal District (TCAD) records anchored to publication date.
- $N = 8$ IRB-approved interviews (ACYY0820) for institutional grounding.

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Research Project Overview

Predicting NIMBYism in Austin's Housing Reform Era

Project Abstract

This research forecasts opposition to development using AI and interprets the mechanics of that opposition in Austin, Texas. By analyzing petition protest letters, public hearing transcripts, and zoning application data, the study evaluates the democratic implications of using predictive analytics to anticipate political behavior. The project employs predictive modeling, natural language processing, and qualitative analysis to predict opposition behavior, identify recurring patterns in opposition rhetoric, and examine their association with planning outcomes.

Study Objectives

Identify Patterns: Characterize the opposition rhetoric and arguments used in protest petitions to oppose rezoning and identify demographic predictors of this opposition.

Evaluate Predictive Tools: Assess whether administrative data and predictive models can accurately forecast opposition to specific development projects.

Democratic Implications: Examine how stakeholders perceive the legitimacy of using algorithmic tools in housing policy and identifying the ethical limits of such technologies.

Participant Role

We are seeking perspectives from urban planners, developers, neighborhood association leaders, and community organizers.

Activity: A 45-60 minute one-on-one interview conducted virtually or in person.

Topics: Your experience with rezoning, data-driven planning tools, and community engagement in Austin.

Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol Y01M00.

Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

Identities will be kept strictly confidential in resulting publications unless you grant explicit permission to be named.

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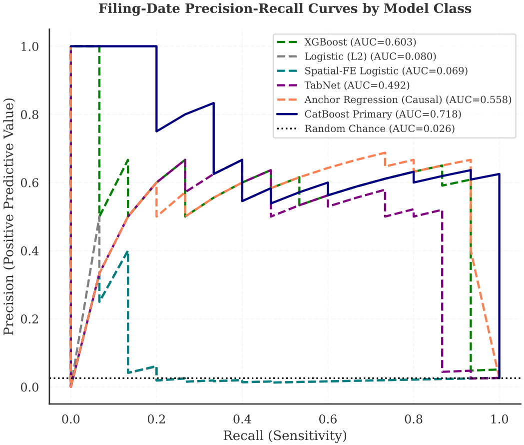
Predicting Resistance: Filing-Date Results

Tree Ensembles Dominate at the Top of the Risk Ranking



Operational Utility: Triage in Practice

Risk Ranking Performance

- Highest-risk 10% of cases: **25.4%** precision.
- **9.9x** lift over the base rate.
- Captures 15 true positives in 59 flagged cases.

Reviewing flagged cases is **nearly 10**$\times$ more likely to surface a genuine petition case.

Appropriate Uses

- ✓ Prioritize outreach for high-risk cases.
- ✓ Flag cases for early negotiation.
- $\times$ Hard probability cutoffs.

Four-Family Modeling Benchmark

Architecture Families Benchmarked

Family	Key Property
Tree (CatBoost)	Best tabular OOD PR-AUC
Deep (TabNet/MLP)	Requires L2; overfits at low N
Linear (L2 LR)	Strong at T+0; decays over time
Robust (V-REx)	Distributional robustness baseline

Selection Criterion

All families tuned to maximize out-of-distribution PR-AUC under strict temporal separation.

Primary model (CatBoost) selected before examining test-set results: best OOD discrimination with the most stable calibration.

Reading the Attribution Charts: Feature Clusters

Cluster	Representative Variables
Parcel Scale	Lot acres, gross site area, deed acreage, lot size (sqft)
Zoning Density	Requested FAR delta, height delta, building coverage delta
Neighborhood Income	Median household income, gross rent, poverty count
Demographics	ACS % White, % Hispanic, % Black, % Asian
Housing Tenure	ACS owner-occupied, renter-occupied, total units
Property Valuation	TCAD appraised value, market value, land value
Historical Protest Activity	Prior protest flag, 1-year and 3-year spatial contagion

Variables with Spearman $|r| > 0.70$ are aggregated into clusters after model fitting; SHAP values are summed within clusters.

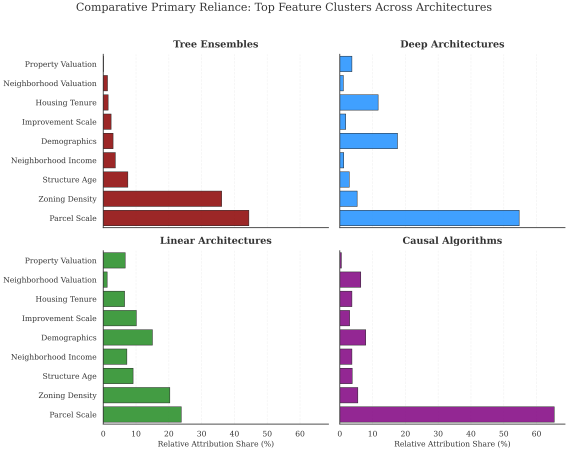
What Drives Petition Risk

Key Finding

Tree ensembles weight the **physical ask**: parcel scale (44%) and zoning density request (36%) together account for 80%+ of attribution.

Deep models also weight parcel scale (55%) but add **neighborhood social fabric**: demographics (18%) and housing tenure (12%).

Same data, same outcome; different internal logic.

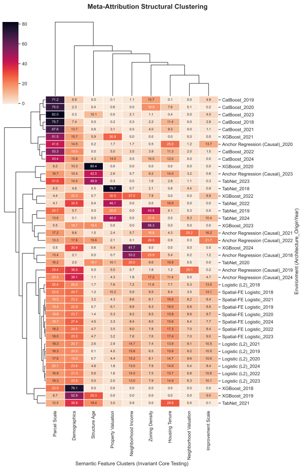


Architecture Determines Feature Reliance

Architecture-Dependent Reliance

- Trees: >80% weight on socio-demographic clusters.
- Deep models: shift toward physical and geographic attributes.
- Same data, same task: different internal logic.

Tree models appear to be detecting opposition to the regulatory ask (scale, density change). Deep models detect opposition from the neighborhood's social composition. These may be different mechanisms of



Temporal Drift and Neighborhood Agency

When Does the Model Break?

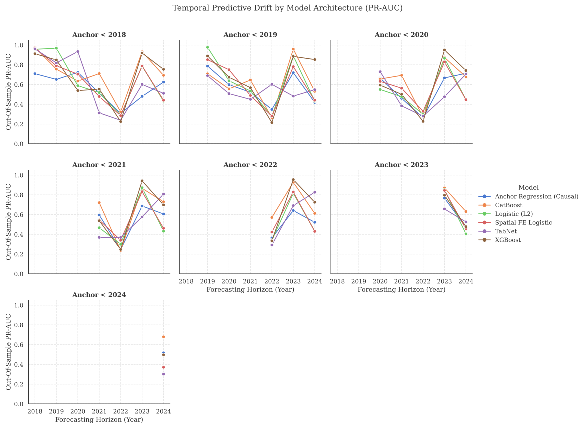
- Linear models collapse at T+3 horizons.
- Tree ensembles degrade more gracefully.

The 2022 Dip

- PR-AUC drops sharply across all families in 2022.
- Consistent with pandemic-era filing disruptions reaching the evaluation window.

Performance sensitivity to ambient shocks is *consistent with* mobilization as a strategic social choice.

Note: descriptive out-of-sample variation; no single institutional break is identified causally.

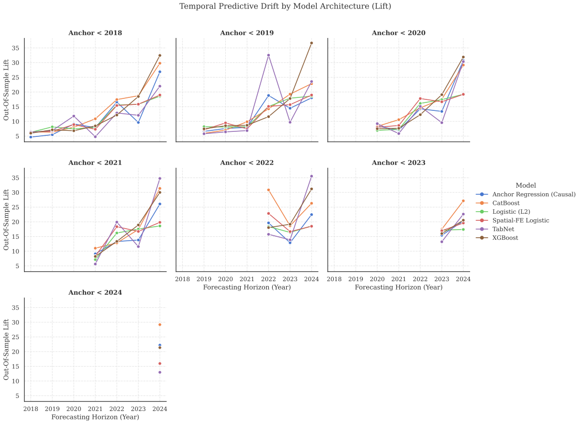


Signal Dilution: When Retraining Hurts

A Counter-Intuitive Finding

- Pre-2018 models achieve $\sim 34\times$ lift in 2024.
- Retraining on modern data degrades lift to $\sim 19\times$.

Post-ordinance labels starve the model of positive examples. The frozen legacy model preserves spatial sorting intelligence from an active institutional era.



What the Algorithm Cannot See

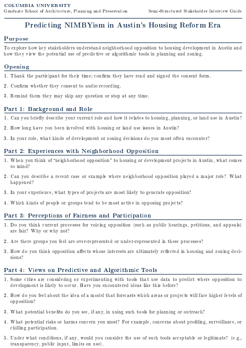
What the Algorithm Cannot Distinguish

- Exclusionary NIMBYism (affluent West Austin blocking density).
- Anti-displacement resistance (East Austin protecting tenants).

Same risk weight; opposite social meaning.

Practitioner Perspectives (IRB ACYY0820, $N = 8$)

- *"Haters are overrepresented and silent supporters are underrepresented."* (Beaudet) $\rightarrow$ explains the model's performance ceiling.
- Rigid cutoffs could miss communities facing real displacement risk. (Torrado) $\rightarrow$ supports triage ranking over hard thresholds.
- Formal opposition data lacks anonymous channels; it is not representative of the full community. (Tomko) $\rightarrow$ the training labels are



COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation — Semi-Structured Stakeholder Interview Guide

Predicting NIMBYism in Austin's Housing Reform Era

Purpose

To explore how key stakeholders understand neighborhood opposition to housing development in Austin and how they view the potential use of predictive or algorithmic tools in planning and zoning.

Opening

1. Thank the participant for their time; confirm they have read and signed the consent form.
2. Confirm whether they consent to audio recording.
3. Remind them they may skip any question or stop at any time.

Part 1: Background and Role

1. Can you briefly describe your current role and how it relates to housing, planning, or land use in Austin?
2. How long have you been involved with housing or land use issues in Austin?
3. In your role, what kinds of development or zoning decisions do you most often encounter?

Part 2: Experiences with Neighborhood Opposition

1. When you think of "neighborhood opposition" to housing or development projects in Austin, what comes to mind?
2. Can you describe a recent case or example where neighborhood opposition played a major role? What happened?
3. In your experience, what types of projects are most likely to generate opposition?
4. Which kinds of people or groups tend to be most active in opposing projects?

Part 3: Perceptions of Fairness and Participation

1. Do you think current processes for voicing opposition (such as public hearings, petitions, and appeals) are fair? Why or why not?
2. Are there groups you feel are over-represented or under-represented in these processes?
3. How do you think opposition affects whose interests are ultimately reflected in housing and zoning decisions?

Part 4: Views on Predictive and Algorithmic Tools

1. Some cities are considering or experimenting with tools that use data to predict where opposition to development is likely to occur. Have you encountered ideas like this before?
2. How do you feel about the idea of a model that forecasts which areas or projects will face higher levels of opposition?
3. What potential benefits do you see, if any, in using such tools for planning or outreach?
4. What potential risks or harms concern you most? For example, concerns about profiling, surveillance, or chilling participation.
5. Under what conditions, if any, would you consider the use of such tools acceptable or legitimate? (e.g., transparency, public input, limits on use).

IRB-Approved Interview Protocol

Limitations

Methodological

- **Chilling effects**: Only filed cases observed; pre-emptive withdrawals are invisible. Selection correction was not feasible.
- **Timestamp imputation**: Milestone dates approximated from statutory minimums.
- **Unobserved certification**: Outcome is reconstructed area share, not the clerk's legal determination.

Ethical and Governance

- **Proxy discrimination**: Neighborhood income and demographics encode class and race dynamics even without protected-class inputs.
- **Motive blindness**: Cannot separate exclusionary NIMBYism from anti-displacement resistance.
- **External validity**: Parameters are Austin-specific and pre-HB 24 only.

Conclusion

What we showed

- Crossing 20% adds 42.5 days of processing time: the outcome is worth predicting.
- Filing-date data ranks petition risk above chance (PR-AUC = 0.718, CI: [0.48, 0.91]), with top-decile lift of 9.9x.
- Algorithm choice determines which feature clusters drive predictions, a normative decision.

What surprised us

- Retraining on modern data degrades lift; legacy models preserve spatial sorting intelligence.
- Performance ceilings are consistent with mobilization as a strategic social choice.

What it implies

- Predictions belong in relative triage rankings, not case-level probability thresholds.
- Methodology is portable; Austin-specific parameters are not.

Implications: A Now-Closed Policy Window

The HB 24 Shift (September 1, 2025)

- Texas HB 24 exempted residential upzonings from the supermajority requirement.
- The 20% institutional veto no longer applies to the most contested case types.
- The institutional regime is closed; its demographic and spatial correlates of opposition remain visible in the predictions.

What travels forward

- Methodology (time-stamped features, temporal holdouts, semantic clustering, qualitative integration) is portable to other discretionary review systems.
- New York, Iowa, Oklahoma, and Arizona retain active petition statutes. The framework applies wherever institutional veto mechanisms create predictable opposition patterns.

Discussion / Q&A

Thank you for your time and feedback.

Dory Kornfeld
Weiping Wu
James Piacentini
Josh Begley

Stijn Van Nieuwerburgh
Matthew Bauer
Adam Vosburgh

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Appendix: Hyperparameter Robustness